Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CSI3005 – Advanced Data Visualization Techniques

Question Bank

Module 1

1. Classify the attributes associated with the following statements as categorical, ordinal, interval and ratio. Justify your classifications with appropriate analysis.
   (i) The company wanted to know how employees perceived the work of upper management. Employees were asked to report the satisfaction of upper management using a 1 to 5 scale (with the following representations: 1 - Extremely Unsatisfied, 2 - Unsatisfied, 3 - Neutral, 4 - Satisfied, 5 - Extremely Satisfied). What type of variable would be considered to represent their response?
   (ii) It was reported that Fridays were generally lighter in terms of number of meetings held. Employees were asked to report the number of scheduled meetings they attended the previous Friday. What type of variable would their response be considered?
   (iii) Horsepower of the motorcycle engines.
   (iv) Temperature of automatic popcorn poppers.

2. Elaborate in detail how task abstraction in data visualization is achieved in terms of actions and targets.
3. A company engages you as a visual idiom designer for New Kids Gaming App. Discuss 3 situations with respect to "KIDS" you will consider for idiom designing. For the identified situations, associate appropriate task, dataset and data type abstractions with justification.
4. Classify the following statements as categorical, ordinal, interval and ratio. Justify your classifications with appropriate analysis.
   (i) Management of a start-up company was playing around with the idea of having a food truck visit the office once a week and was trying to gauge how much employees would spend to help entice various food truck owners. Employees were asked to report the amount of money they believe they would spend on lunch (in $YY.YY) if a food truck came to the office once a week. What type of variable would their response be considered?
   (ii) Number of cartons of milk manufactured every month.
   (iii) Temperatures of Liquid Barium Oxalate.
   (iv) Students rank awarded by the University.
5. State the need of validation. Explain in detail various threats and validation approaches in four levels of visualization design.
6. Consider healthcare industry for data visualization scenario and investigate the chosen scenario meticulously based on data and task abstraction factors for an effective presentation to the customer with suitable justifications.
7. Explicate different glyph techniques used to visualize vector fields. And discuss the issues in each technique.

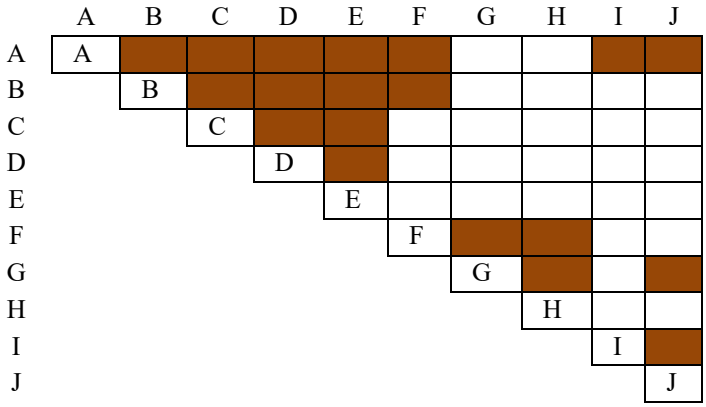
Module 2

8. Elaborate on color coding techniques used in X-Ray and CT-scan of human organs.
9. The scalar values for 10x10 are given below

55	55	55	55	55	55	55	55	55	55
51	50	45	38	36	36	38	40	42	47	50
51	49	30	20	29	31	26	30	33	40	50
51	42	25	15	28	25	21	20	23	27	50
51	32	19	23	26	21	16	10	12	24	50
51	32	30	33	25	22	12	02	09	21	50
51	36	35	34	26	18	15	14	13	25	50
51	39	36	35	30	27	20	20	23	29	50
51	45	39	36	35	33	30	29	30	48	50
51	50	50	50	50	50	50	50	50	50	50

Construct an iosline for the above 2D dataset for V=5, V=11, V=18, V=22, V=28 and V=37.

10. The network adjacency matrix for a specified network is given below, visual the network idioms with appropriate color. Identify the structures of the network visible in the visual idioms and explain.



11. a) With a neat idiom explain different color palettes in Univariate. (5)
    b) With a neat sketch explain different colormaps in bivariate. (5)

12. Design bivariate color map for the *diamonds* dataset (*cut vs color*).

		D	E	F	G	H	I	J
Fair	5	6	7	4	5	4	3	
Good	18	23	11	16	17	7	7	
Very Good	26	37	46	40	33	25	13	
Premium	34	53	43	49	41	24	13	
Ideal		54	63	79	92	53	34	15

Assume the color 'F' in the *diamonds* dataset and apply the color map to encode its various cut from *Fair* to *Ideal*.

13. Explain and compare various Tree drawing idioms.

Module 3

14. Consider a *Healthcare analytics* dataset, explain the steps and create a dashboard-using tableau with proper idioms.
15. Consider an *E-Commerce* dataset and create a dashboard using shiny application in R Code.
16. Consider a 'Data Visualization' book with 10 chapters. The entire text book contents are available in the file '*books.rda'*. Using R code
    i. convert all non-tidy text to tidy text.
    ii. compute total number of positive and negative words.
    iii. Plot top 10 positive words and negative words in decending order.
    iv. Use appoprate idioms to display the top 25 negative words.
17. The airquality dataset with the dimension 153X6 is shown below

Ozone	Solar.R	Wind	Temp	Month	Day	
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
…	…		…		…	
151	NA	191	14.3	75	9	28
152	18	131	8.0	76	9	29
153	20	223	11.5	68	9	30

Using R code with proper visual idioms display the gird chart with 2X3 consists of scatter plot, line plot with point and line, vertical bar plot, horizontal bar plot, box plot and 3D Pie chart

18. Consider a wellness center dataset that contains attributes like CityName, CityCode, WellnessCenterCode, WellnesscenterName, WellnessCenterAddress, ContactNumber, NumberofDoctors, Type (Allopathic/ Unani/ homeopathic/ Ayurvedhic), NumberofBeds. Write a suitable R function for the following scenarios
    a. Display the CityName and CityCode which has Number of Doctors >3 and Type = Allopathic.
    b. Filter the Wellness centers that have NumberofBeds > 10 and group it by CityCode and display the result for each City with NumberofBeds and NumberofDoctors .
    c. Find the rows with no missing value for Contact Number.
    d. Write a function to filter the rows where Wellness Center Address and Contact Number are available. Group the data by CityCode and Type. Finally, calculate the total NumberofDoctors employed for each group.
19. In tableau, write step by step procedure to visualize the commodity monthly price data for the years 2020-2023 to the scenarios given below.

a. Connect to a commodity_priceindex excelfile from your system and load to tableau.
b. Describe how the datatypes are assigned and categarized for the attributes commodityname, commoditycode, commodityweight, price012020, price022020…price112023 price122023 by tableau
c. To Compare the commodities with prices >200 for January 2021, January 2022 and January 2023, choose two appropriate visualization techniques
d. Show the attributes considered for rows and columns in each of the chosen techniques with appropriate marks and filters

Module 4

20. Compare and contrast area mark based geospatial data visualization Techniques.
21. Write R code to visualize population of metropolitan cities and states of India using appropriate geospatial visualization techniques. Include comments to clarify the usage of the statements.
22. Define a choropleth map and explain two common use cases in geospatial data analysis (for example, in demographics or public health).
23. Explain the concept of hexagonal binning in the context of spatial point data, and contrast it with simple point maps.
24. Define a cartogram and explain how it differs from a choropleth map in terms of encoding the data variable.
25. A city government has shared three datasets:
    a. Polygon boundaries of wards,
    b. Point locations of reported traffic accidents, and
    c. Population data for each ward.

    Design a geospatial visualization dashboard in R to help identify high-risk areas and patterns. a) Propose an appropriate combination of at least three of the following map types: choropleth map, hexbin map, dot map, cluster map, and cartogram map. Justify briefly why each chosen map type is suitable for a specific analytical question. b) Outline a high-level R workflow (packages and main steps) to implement your proposed solution, mentioning how you would handle data loading, spatial joins, and map visualization.

Module 5

26. The River Nile annual streamflow data for the year 1871 to 1970 sample is given below

Time Series: Start = 1871 End = 1970 Frequency = 1
[1] 1120 1160 963 1210 1160 1160 813 1230 1370 1140 995 935 … … ….
Using R code to build a suitable model to predict for the year between 1971 and 1980. (Use the suitable idioms to display the output for each steps)

27. The total observations in the diamonds dataset are 53040x10. The sample dataset are given below.

carat	cut	color	clarity	depth	table	price	x	y	z

	<dbl>	<ord>	<ord>	<ord>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	0.23	Ideal	E	SI2	61.5	55	326	3.95	3.98	2.43
2	0.21	Premium	E	SI1	59.8	61	326	3.89	3.84	2.31
3	0.23	Good	E	VS1	56.9	65	327	4.05	4.07	2.31

From the given observation visualize any 5 idioms for multivariate analysis.

28. Compare and contrast two common techniques for visualizing multivariate data: 2D Scatterplots and Parallel coordinates. For each technique, identify the strengths and weaknesses in terms of the three following basic visualization tasks:
    (i) Find value of data case
    (ii) Find correlation
    (iii) Find outlier
29. What are the different visual techniques available for summarizing a text document, understanding the sentiments in tweets and finding the topics in a document? Explain briefly with examples and visual presentations.
30. Consider a 'Advanced Data Visualization Techniques' text book with 15 chapters. The entire text book contents are available in the file 'books.dat'. Using R code
    a) convert all non-tidy text to tidy text.
    b) compute total number of positive and negative words.
    c) plot top 10 positive words and negative words in descending order.
    d) use appropriate idioms to display the top 25 negative words.

Module 6

31. Describe two real-world applications where data streaming is preferred over batch processing, clearly stating the reason in each case.
32. Consider a clickstream from an e-commerce website. Describe how a streaming system could process this data in real time to compute "number of clicks per user per 5-minute window," mentioning the main logical steps.
33. Elaborate on the integration process of incorporating real-time data streams into various analytical tools.
34. A financial institution wants to detect fraudulent credit card transactions in real time using streaming data.
    a) Explain the term streaming analytics and describe how it supports low-latency decision-making.
    b) Propose a high-level architecture for this use case, naming the main logical components (data ingestion, stream processing/analytics, storage, and visualization/alerting).
    c) Describe two analytical techniques that can be applied on the data stream (for example, rule-based detection, statistical thresholds, or machine learning models), and explain how their results can be integrated into a real-time visualization for analysts.

Module 7

35. ABC Health Insurance Company has been facing challenges in understanding and optimizing their claims processing system. They have collected extensive data over the years related to claims, including patient demographics, diagnosis codes, treatment procedures, claim amounts, and processing times. However, they lack a comprehensive understanding of their data and struggle to identify patterns, trends, and potential areas for improvement. As a data analyst consultant, you have been hired to assist ABC Health Insurance Company in leveraging their data effectively to enhance their claims processing system and overall operational efficiency.
    a) Propose a strategy for abstracting and organizing the data to facilitate analysis and decision-making.
    b) Define the specific tasks or objectives that ABC Health Insurance Company aims to accomplish through data analysis and optimization of their claims processing system.
    c) Choose four validation methods from the following options and explain how each method can be applied to ensure the accuracy, reliability, and validity of the
    findings derived from the data analysis.
    d) Based on the identified data variables and objectives, recommend suitable visualization tools for effectively visualizing the data and communicating key findings.
    e) Justify your choice of visualization tools based on their ability to highlight patterns, trends, and anomalies in the data relevant to claims processing optimization